

Introduction

Good afternoon everyone. Today, I will be demonstrating **CarOneStop**, our mobile application that helps users search for car parks and EV charging stations, receive recommendations based on their preferences, navigate to their destination, manage wallet top-ups, and perform EV booking actions. In this demo, I will walk through the full user journey in one continuous flow, starting from authentication, then moving into map search and recommendation features, before ending with the EV booking lifecycle, including cancellation and check-in.

Authentication: Login

I will begin at the login screen using a pre-verified account so that I can quickly enter the main system. Here, the user enters their credentials and taps **Login**. This demonstrates the standard authentication flow, where the credentials are validated by the backend and, upon success, the user is granted access to the main application. Once login succeeds, the user is brought directly to the home map screen, which serves as the entry point for the core features of CarOneStop.

Authentication: Registration and Verification

Next, I will log out and return to the login page so that I can demonstrate the registration process. From here, I tap **Register** and fill in a valid set of registration details, including email, username, password, car type, car model, and charging type. This is important because these profile details are not only used for account creation, but also support later EV-related logic such as compatibility checking. After submitting the form, the system redirects the user to the **Validate Email** screen. At this point, I can either enter the OTP directly or tap **Resend OTP** to show that the system supports regeneration of a verification code. Once the correct OTP is entered, the account moves from the unverified state into a verified state, making it fully usable for subsequent login and application features.

Authentication: Forgot Password and Reset Password

After showing registration, I will demonstrate the password recovery flow. From the login screen, I tap **Forgot Password**, enter the email address, and submit the request. This triggers the tokenized password reset process. Once the reset link is received, I proceed to the reset-password page and enter a new password. This demonstrates that the system supports secure account recovery through a token-based reset mechanism. After the reset is completed successfully, the user can return to the login page and sign in again using the updated password.

Home Screen and Map Search

With authentication complete, I now log in again and arrive at the home map screen. This is where the main location-based features of CarOneStop are accessed. To demonstrate search, I type a location such as **Suntec City** into the search bar and select one of the suggestions. The system retrieves the location, places a marker, and re-centers the map to the selected destination. This shows how CarOneStop supports live destination search and map-based interaction, allowing users to quickly focus on an area of interest before exploring nearby facilities.

Carpark Recommendation Features

Once the map is centered on the selected destination, I switch to the carpark feature and demonstrate the recommendation modes. I first set an expected parking duration, because duration affects effective price calculation in the recommendation logic. Then I cycle through the main preference modes: **Cheapest**, **Nearest**, **Most Available**, and finally **Custom**. In custom mode, I adjust the sliders to reweight cost, distance, and availability. As I do this, I explain that the recommendation system changes how nearby carparks are ranked based on the selected preference, showing how users can either rely on preset strategies or fine-tune the recommendation process according to their own priorities.

Navigation

After selecting a suitable carpark, I demonstrate the navigation feature by tapping **Start Navigation**. This changes the application into a navigation view where the user can see the route overlay together with route guidance information such as ETA and directions. I briefly show the navigation interface, then tap **Stop Navigation** to return to the normal map screen. This demonstrates that the application not only recommends facilities, but also supports the user in actually reaching the chosen destination.

Wallet Top-Up and Transaction History

Next, I move to the **Wallet** tab to demonstrate the payment flow. Here, I choose a predefined top-up amount and continue through the payment flow, either through a card-style input or QR-based process depending on what is available in the interface. Once the top-up is confirmed, the wallet balance increases immediately. I then open the **Transactions** tab to show that the top-up has been recorded in the user's transaction history. This demonstrates both the wallet balance management feature and the traceability of transactions within the system.

EV Preferences and EV Station Discovery

After showing wallet functionality, I go to the **Profile** page and briefly confirm that the EV-related settings, such as car type and charging type, are correctly stored in the user profile. I explain that these settings are important because they influence EV charging compatibility later in the booking flow. Returning to the home screen, I switch to the **EV Stations** feature so that the map now displays EV charging station markers instead of carparks. This shows that CarOneStop supports both ordinary carpark discovery and EV-specific discovery, while also tying the EV charging flow to the user's saved vehicle and charging profile.

EV Booking Flow

I then select an EV station marker and proceed with the EV booking flow. I choose a compatible plug type and tap **Book Charger**, then confirm the booking in the modal. Here, I explain that the system enforces important booking rules such as EV-only eligibility, compatibility with the saved charging type, and active booking constraints. If all conditions are satisfied, the booking is created successfully, the booking fee is deducted from the wallet, and the user receives a booking result with the reservation details. This demonstrates the most important transactional flow in the EV charging feature.

Booking Cancellation

After completing one booking, I go to the **Transactions** tab and open the active booking card. From here, I tap **Cancel** and confirm the action. This demonstrates the cancellation flow and the change in booking state from an active reservation to a cancelled booking. It also shows that booking actions remain visible and manageable after the original booking has been created.

Booking Check-In

Finally, I created one more quick EV booking so that I can demonstrate the check-in process. After the booking is created, I go again to the booking section in **Transactions** and tap **I'm Here**, which takes me to the check-in page. I then simulate the expected completion action, such as scanning or code submission, depending on the interface available. Once completed, the system shows a success or completion modal, confirming that the user has successfully checked in to the reservation. This closes the loop on the full EV booking lifecycle, from discovery to reservation, post-booking action, and check-in completion.

Conclusion

To conclude, this demonstration has shown the complete end-to-end flow of CarOneStop. I started with authentication, covering login, registration, email verification, and password reset. I then moved into the map-based experience with location search, recommendation logic, and navigation. After that, I demonstrated wallet top-up and transaction traceability, before showing how EV preferences influence charger compatibility and how the EV booking lifecycle supports booking, cancellation, and check-in. Together, these features illustrate how CarOneStop integrates parking, EV charging, navigation, and payment into a single coherent mobile platform.